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## No Bird in Charge

Student copy (project or hand out)

- 1 At dusk in late autumn the starlings gather above their roost and begin to move as one body. There may be a thousand of them, or fifty thousand. The flock climbs, folds, pours sideways, thins to a gray ribbon, knots into a fist. It has no leader. No single bird decides the shape, and none is told it.
- 2 How a crowd that large turns without colliding was a real puzzle until recently. The current answer is that each bird is not watching the whole flock at all, but only its handful of nearest neighbors, about seven of them, matching their speed and heading whatever the distance between them. A turn begun at one edge passes inward, neighbor to neighbor, so a ripple of banking can cross the entire flock faster than any one starling flies. The order is real, and it is built entirely from the bottom.
- 3 The effect is sharpest when a predator arrives. A peregrine falcon stoops at the edge, and the flock answers not by scattering but by tightening, the threatened side pulling inward so fast it darkens like a bruise spreading under skin. Thousands of separate animals, and not one collision; total panic, and perfect order. The falcon, built to pick a single target, loses the single target in the boil of bodies.
- 4 Why they do it at all is not fully settled. The safety of the crowd is part of it, surely; but researchers have also tied the evening gatherings to warmth, to the trading of word about where the food was that day, to the simple pull of a good roost, and to reasons no one has yet named. The murmuration is not a decision and not a message. It is what happens when ten thousand small rules, each followed by a bird watching only its neighbors, add up in the cold air over a winter field.
- 5 From the ground it looks like a single mind choosing where to go. There is no mind, and there is nothing choosing; there is only each bird, and its seven neighbors, and the dusk coming on.

### AP-style multiple choice:

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1. In the third paragraph, the comparison of the tightening flock to 'a bruise spreading under skin' primarily serves to
  - A) suggest that the flock has been injured by the diving falcon
  - B) give the flock's sudden, spreading reaction a precise and physical image
  - C) register the writer's sympathy for the hunted birds
  - D) compare the starlings unfavorably with their predator

2. The writer follows the long sentence describing the flock's motion with the short sentence 'It has no leader.' This juxtaposition primarily functions to

- A) stress that the flock's intricate order arises without anything directing it
- B) signal that the writer will next explain how a leader is chosen
- C) concede that the flock's movements are ultimately random
- D) summarize a scientific debate introduced earlier in the passage

### **Discussion questions:**

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1. The writer describes the murmuration almost entirely through precise physical detail and never once calls it beautiful or amazing. How does withholding those words change the way the wonder of the scene reaches the reader?
2. Twice the writer admits the science is unsettled ('was a real puzzle until recently,' 'reasons no one has yet named'). How does admitting what is not known affect your trust in the writer as an observer?

### **Writing prompt (homework or extended in-class, Q2 rhetorical analysis):**

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Write an essay analyzing how the writer uses imagery, sentence variety, and a restrained, observational tone to convey the wonder and the order of a starling murmuration without ever asserting that it is wondrous.